

Vol 11 Chapter 3 — Wallach Representations and K-Type Theory

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 3 — Wallach Representations and K-Type Theory

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Wallach 1976 (L1 ESTABLISHED BST source) constructed an explicit family of unitary representations of $SO_0(p, 2)$ supported on bounded symmetric domains D_{IV}^p , organized into layers — for $p = 5$, the substrate has 5 Wallach layers (matching BST primary $n_C = 5$), and these layers anchor BST's particle mass hierarchy via “mass hierarchy from Wallach layer index” (Vol 2 Ch 6).

Level 2 — graduate-physicist precision

3.1 The Wallach 1976 construction

Nolan Wallach (1976) constructed an explicit family of unitary irreducible representations of $SO_0(p, q)$ supported on bounded symmetric domains, generalizing the holomorphic discrete series.

For $SO_0(p, 2)$: representations parameterized by an integer parameter k + auxiliary discrete data. The layers $k = 0, 1, 2, \dots, p - 1$ form the Wallach layer structure.

3.2 5 Wallach layers for D_{IV}^5

For BST $SO_0(5, 2)$: $p = 5$, so 5 Wallach layers indexed $k = 0, 1, 2, 3, 4$.

The layers carry different unitary structures: - Layer 0: trivial rep - Layer 1, 2, ..., 4: various unitary modules supported on D_{IV}^5

3.3 K-type organization

Each Wallach representation is realized on a Bergman-type space; its K-type decomposition (restriction to $K = SO(5) \times SO(2)$) organizes the substrate's natural state spectrum.

For the substrate's bottom Wallach layer: K-types include the trivial K-type (ground state, Vol 5 Ch 1) and the (1, 1) K-type with Casimir = $C_2 = 6$ (first excited).

Each Wallach layer adds new K-types not present in lower layers — building up the substrate's K-type spectrum.

3.4 Mass hierarchy via Wallach layer index

Vol 2 Chapter 6 (“mass hierarchy from Wallach layer index”; BST Task #60 completed): the 3 generations of fermions (e, μ , τ ; u/d, c/s, t/b) are positioned in different Wallach layers — the layer index controls the mass hierarchy.

This connects 5 Wallach layers (substrate-mathematical) to 3 generations (observed particle physics) via a specific BST mechanism.

3.5 Casey's BST primary $n_C = 5$

The number of Wallach layers = $p = 5$ in $SO_0(5, 2) = \text{BST primary } n_C$. This is not coincidence in BST framework — $n_C = 5$ IS the Wallach layer count, by substrate's natural construction.

The BST primary n_C controls: - Number of Wallach layers (5) - Orbital degeneracy $2\ell+1$ at $\ell=2$ (= 5) - Substrate cyclotomic period (Vol 6 Ch 9 RG cascade) - Speaking-pair period in heat kernel arithmetic triangle (Paper #9) - Hilbert polynomial $P(\mathbb{Q}^5, m = 2) = N_C^3 = 27$ via 27a1 / $N_c \cdot n_C$ structure

3.6 Connection to particle multiplets

The substrate's $SO(5)$ K-type structure $\times$ Wallach layer organization is what BST uses to organize the Standard Model particle multiplets (Vol 2). The specific multiplet structure of quarks/leptons is the substrate's natural Wallach K-type assignment.

3.7 K-audit anchors

- Wallach 1976 (L1 ESTABLISHED): the foundational reference
- Task #60 completed: Wallach layer index $\rightarrow$ mass hierarchy
- Vol 2 Ch 6: explicit mass hierarchy derivation

Level 3 — 5th-grader accessibility

Wallach 1976 constructed unitary representations of $SO_0(p, 2)$ on bounded symmetric domains. For BST's $SO_0(5, 2)$: 5 Wallach layers (indexed $k = 0, 1, 2, 3, 4$).

The 5 = BST primary n_C . Each layer has its own K-type spectrum. 3 generations of fermions (electron/muon/tau, etc.) sit at different Wallach layers $\rightarrow$ mass hierarchy. This is “mass hierarchy from Wallach layer index” (Vol 2 Ch 6).

What comes next

Chapter 4 develops holomorphic discrete series and highest weights.

Where to look this up

- Wallach 1976, “Analytic continuation of the discrete series” Trans AMS
- BST Task #60; Vol 2 Ch 6